

FINAL SEMINAR · OCTOBER 2026

Multi-Page Visually Rich Document Understanding

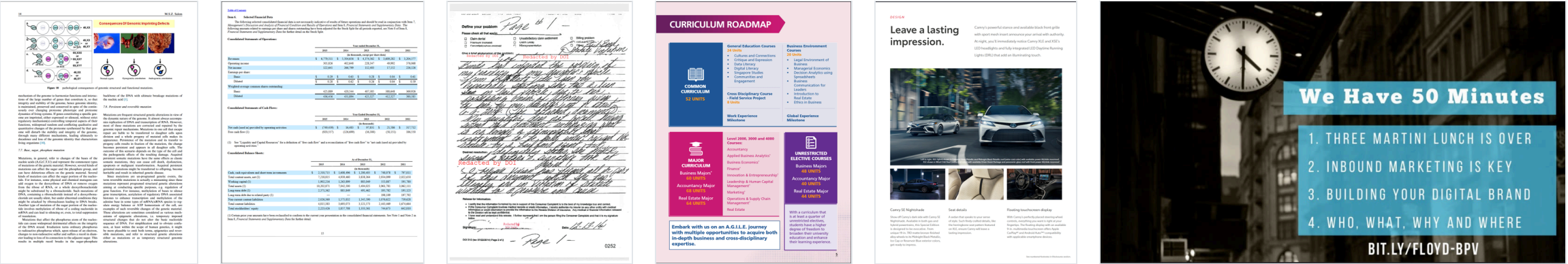
A survey of system design and an empirical attribution of failure

Lewei Xu

Supervised by Dr. Yihao Ding · Dr. Daochang Liu

Visually Rich Document Understanding

A visually rich document is one whose meaning is carried jointly by text, layout, and visual content rather than by text alone. Understanding one is commonly framed as **answering questions over it**.



Representative pages from MMLongBench-Doc

- Text is small and dense. Legible only at far higher input resolution than a typical vision-language task requires.
- Layout contains semantic information. A number means one thing in a table's first column and another in its second.
- Reading order must be recovered. A page is a two-dimensional arrangement rather than a linear stream.
- "Which year had the largest gap between projected and actual spending?"
- "How many of the diagrams in this manual carry a safety warning?"
- "What does the appendix table referenced in Section 3.2 report for the second condition?"
- "What was the total headcount at the end of the year?"

General-Domain MLLMs and VRDU

VRDU long predates general-domain MLLMs; what has changed is that those models now sit at the centre of it. Closed-source assistants have steadily absorbed VRDU techniques into their products, while open-weight models carry the same techniques to documents that cannot leave the premises.

Closed-source products

Three routes cover most of what a consumer assistant does with an uploaded document.

- **Read it whole.** Parse the PDF for text, render it to per-page images, and place both in the context window.
- **Read it on demand.** The file stays on disk, and tool calls extract or search particular pages.
- **Retrieve over it.** Content beyond the context window is indexed and searched rather than ingested.

ChatGPT · Claude · Gemini · NotebookLM

Open-weight models, on premises

Much document work involves material that cannot be sent to a third party at all.

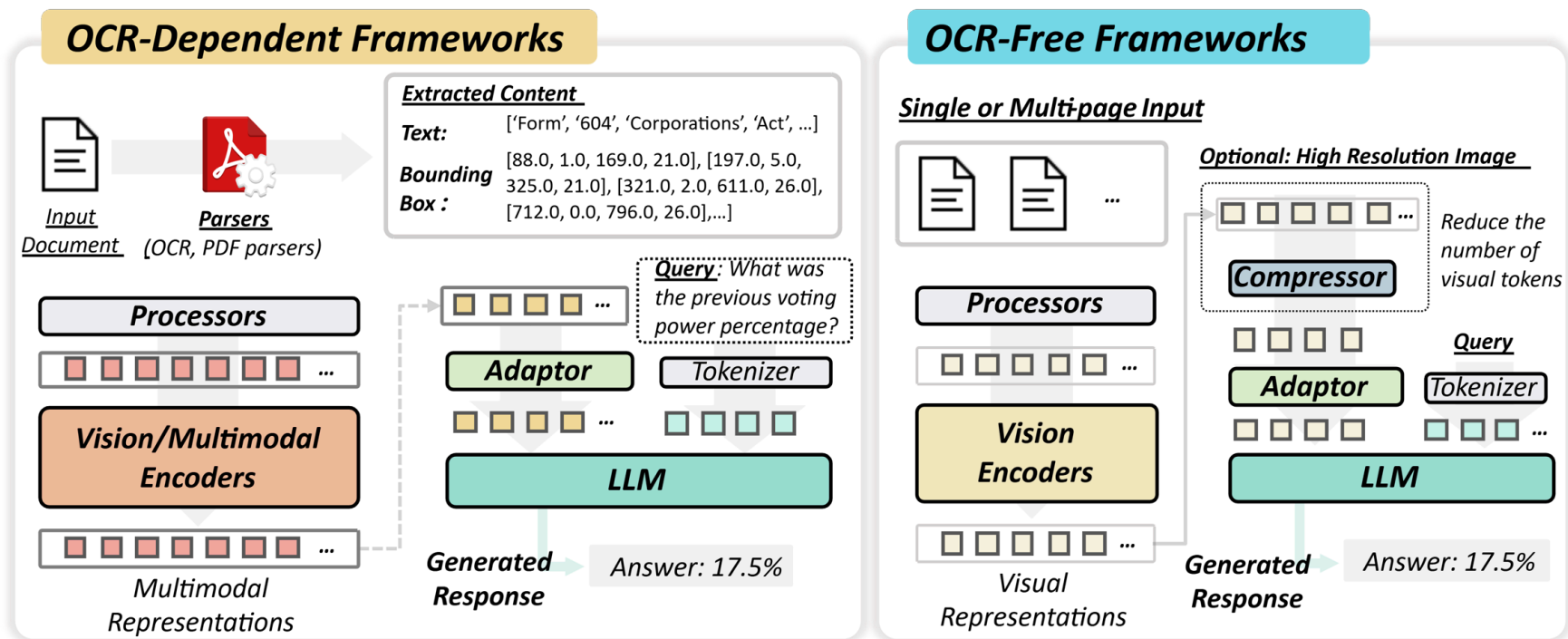
- **Documents that cannot leave.** Medical records, legal and financial filings, and mining and resources data.
- **The same pipelines, local weights.** Open-weight backbones rebuilt behind the firewall rather than called over an API.
- **Hardware is the binding constraint.** Quantisation, retrieval, and page selection do the work the context window cannot.

Qwen3-VL · Gemma · InternVL

Both are document-understanding pipelines. The frontier is how these backbones can be leveraged, together with the machinery built around them, to answer questions over documents they **cannot process directly**.

Single-Page VRDU Overview

The single-page setting is largely solved, and multi-page systems inherit their per-page machinery from it.



OCR-dependent and OCR-free architectures for MLLM-based VRDU. Adapted from Ding et al.

OCR-dependent

Parse the page first, recovering text with its bounding-box coordinates. Captures structure without heavy visual pretraining, but ties accuracy to the parser's.

OCR-free

A vision encoder reads the rendered page from pixels. Handles handwriting and irregular layouts, but needs high resolution, so compression usually follows.

Challenges of the Multi-Page Setting

- **The document exceeds the context window.** Long multi-page documents routinely outrun what a backbone can hold, so the system must select which parts of the document reach the MLLM at all.
- **Scaling the context does not resolve it.** Long-context models attend poorly to evidence buried mid-input, and memory and compute grow with sequence length, so the achievable context is bounded by hardware.
- **Every required evidence unit must be located.** Answering often means integrating evidence from several parts of the document at once, and missing any one of them is enough to answer incorrectly or not at all.
- **Cross-page structure is not recoverable in isolation.** Section hierarchy and cross-references cannot be read off any single part of the document, yet questions often need them to gather the evidence in the first place.

Seminar Overview

Two papers came out of this Honours project. The rest of the seminar follows them in turn.

1 **Managing Evidence at Document Scale: A Survey of Multi-Page Visually Rich Document Understanding**

Lewei Xu, Yihao Ding, Zihan Xu, Siwen Luo, Daochang Liu, Wei Liu

Main Conference of EMNLP 2026

NEXT MP-VRDU organised through evidence management: a taxonomy of four architectural families, multimodal representation, and inference-time and training strategies.

2 **Locating Failure in Multi-Page Visually Rich Document Understanding: An Empirical Attribution**

Lewei Xu, Yihao Ding, Zihan Xu, Daniel Yitian Su, Daochang Liu, Siwen Luo, Yifan Peng, Wei Liu

Submitted to EACL, under review

THEN Failure attributed to three loci, representation, selection, and reasoning, through controlled interventions on a single pipeline.